

Particle contamination in the microfabrication

contamination: sự làm nhiễm bẩn

microfabrication:

I. Introduction

A particle is defined as a solid or liquid entity. These particles can range in size from microscopic to visible. These particles can be categorized as either natural, e.g. dust or pollen, or synthetic, e.g. micro-plastics or synthetic polymers. Despite their minute size, these particles can exert a considerable influence on micro and nano fabrication processes, such as the miniaturization of chip devices, the interaction of electric fields, thermal effects, chemical reactions, surface contamination, and manufacturing processes [1]-[6].

For micro or nano-scale chip devices, even a small particle or impurity on the surface of a semiconductor can significantly affect its function. Contamination has been shown to disrupt the precise patterns and structures within the semiconductor, leading to defects, electrical malfunctions, or even complete failure [1], [3].

Semiconductor devices are dependent on precise electrical properties for optimal functionality, including reduced signal quality, increased power consumption, and the potential for short circuits. The presence of contaminants can modify these electrical properties by interfering with the electric field within the semiconductor [1], [2]. The presence of even a minute particle or impurity has the potential to generate a local electric field that deviates from the desired behavior, thereby affecting the performance and reliability of the semiconductor device.

It is imperative to note that semiconductor devices generate heat during operation, and effective heat dissipation is paramount to ensure optimal performance and reliability [1]. Contamination, such as dust or foreign particles, has been shown to impede heat transfer or act as an insulator, resulting in elevated temperatures in localized areas [2]. Such elevated temperatures have been shown to have a detrimental effect on the performance of semiconductor devices, to alter their electrical characteristics, and to potentially cause thermal damage or device failure.

Semiconductors are susceptible to chemical reactions with certain contaminants, resulting in the formation of undesirable or corrosive compounds [1]. These chemical reactions can lead to alterations in the material's properties, conductivity, impurity levels, or structural

defects, consequently reducing the performance of the semiconductor device or causing failure.

The surface of a semiconductor device is particularly susceptible to contamination by particles, such as dust, oil, or residue, which can accumulate and form barriers or insulators that interfere with electrical connections or impede heat dissipation [1]. Surface contamination has the capacity to affect bonding processes, modify adhesion properties, or introduce defects into the layers or thin films of a semiconductor device, thereby reducing the functionality of the device.

Finally, it is important to note that semiconductor fabrication involves a number of complex manufacturing processes, such as photolithography, deposition, etching, and wafer processing [3]. Contamination has the potential to occur at any stage of the process. Even minor deviations in process parameters, improper handling practices, or inadequate housekeeping controls can serve as potential sources of contamination. The consequences of contamination at any stage of the manufacturing process can be significant, potentially resulting in device failure, diminished productivity, or compromised device performance.

The size of the particles and the number of particles can cause defects in the process or chip, potentially leading to failure [2]. The semiconductor process may be subject to four kinds of defect: bulk silicon or crystalline defects, patterning defects, particle defects and workmanship defects. These defects have the potential to compromise the integrity of semiconductor devices. Consequently, it is imperative to implement stringent measures to avert particle contamination during the semiconductor manufacturing process. Despite the implementation of stringent cleanroom protocols, including ventilation, airflow management, and filtration systems, contamination can still occur during various stages of the manufacturing process, particularly during the etching or deposition steps. To ascertain the level of contamination in a given process, engineers frequently utilize the optical microscope.

In order to evaluate the contamination level of the process, engineers frequently utilize the optical microscope. These devices are often situated within the lithography, or coating/development rooms, where the utilization of yellow light is standard practice to prevent unwanted resist exposure and development. The majority of photoresists are only sensitive to light with a higher frequency than yellow light, namely blue and UV light.

Mercury lamps emit a number of discrete frequencies of light that are utilized in this context. The wavelength of light is typically measured in nanometers (nm). The G-line is 436 nm, and is blue in color. I-line, on the other hand, has a wavelength of 365 nm and is classified as a UV light. The wavelength of yellow light ranges from 560 to 590 nm, and it is important to note that photoresist exhibits no sensitivity to this particular range [7]. However, this yellow light can impede the ability of the technicians to discern the optical image. This hindrance can be exemplified by the difficulties engineers encounter in quantifying the number of particles, the size variations within contaminated areas, and the dimensions of the particles themselves. Moreover, the time available for the technician to observe in the cleaning room is limited. The analysis and identification time is contingent on the technician's level of experience. It is important to note that technicians typically provide only an overview of the image, which inevitably results in the loss of crucial details when examining the specimen under the optical microscope. The ability to distinguish particles based on their physical characteristics is a key skill in the engineer's toolkit. The utilization of artificial intelligence (AI) has been identified as a significant support for the processes of learning and detecting particle contamination, analyzing time as well as evaluating the contamination level of a semiconductor wafer.

A number of studies have been conducted in the field of semiconductor manufacturing environment with regard to deep learning research. These include the use of convolutional neural networks for image computing [8], the development of deep learning-based computer vision algorithms for 2D materials [9], and distance metric learning for the detection and classification of noisy optical images [10].

II. Methods

a. Data Acquisition

The preparation of the dataset entailed the collection of 50 images. The Boron GaN samples with different thickness were fabricated via the sputtering deposition method. A sputtering target is defined as a source of contamination in the deposition system. This is due to the generation of gaseous, vapour, or particulate matter during the sputtering process. Such contamination can arise from second phase particles in the target material. Also, this step makes contaminations due to the uniform bombardment from the target XXX. These

layers were deposited on silicon wafer Si. The process always needs annealing after the deposition process. However, it leads to impurity layer appearing more clearly. The sample images were captured using ??? (optical microscope)

- b. Data labeling
- c. Model and training

III. Results

IV. Conclusions

Reference

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